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TRAFFIC BASED DRIVING ANALYSIS

Abstract

A driving analysis server may be configured to receive vehicle operation data from vehicle sensors and telematics devices of a first vehicle, and may use the data to identify a potentially high-risk or unsafe driving behavior by the first vehicle. The driving analysis server also may retrieve corresponding vehicle operation data from one or more other vehicles, and may compare the potentially high-risk or unsafe driving behavior of the first vehicle to corresponding driving behaviors in the other vehicles. A driver score for the first vehicle may be calculated or adjusted based on the comparison of the driving behavior in the first vehicle to the corresponding driving behaviors in the other vehicles.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of and claims priority to co-pending U.S. patent application Ser. No. 17/318,561, filed May 12, 2021, and entitled “Traffic Based Driving Analysis,” which is a continuation of and claims priority to U.S. patent application Ser. No. 15/985,801, filed May 22, 2018, and entitled “Traffic Based Driving Analysis,” which is a continuation of and claims priority to U.S. application Ser. No. 15/375,967 (now U.S. Pat. No. 10,005,471), filed Dec. 12, 2016 and entitled “Traffic Based Driving Analysis,” which is a continuation of and claims priority to U.S. application Ser. No. 14/731,745 (now U.S. Pat. No. 9,558,656), filed Jun. 5, 2015, and entitled “Traffic Based Driving Analysis,” which is a continuation of and claims priority to U.S. patent application Ser. No. 13/770,622 (now U.S. Pat. No. 9,081,650), filed Feb. 19, 2013 and entitled “Traffic Based Driving Analysis,” which is a non-provisional of and claims priority to U.S. Provisional Application No. 61/739,439, entitled “Traffic Based Driving Analysis,” filed Dec. 19, 2012. Each of these applications is incorporated by reference in its entirety herein.

TECHNICAL FIELD

[0002] Aspects of the disclosure generally relate to the analysis of driving data and calculation of driver scores. In particular, various aspects of the disclosure include a framework for evaluating a driving behavior at a vehicle using vehicle sensor data and telematics from a plurality of other vehicles and other data sources.

BACKGROUND

[0003] Telematics includes the use of technology to communicate information from one location to another. Telematics has been used for various applications, including for the exchange of information with electronic sensors. As telematics technology has progressed, various communication methodologies have been incorporated into automobiles and other types of vehicles.

[0004] Telematics systems such as on-board diagnostics (OBD) systems may be used in automobiles and other vehicles. OBD systems may provide information from the vehicle's on-board computers and sensors, allowing users to monitor a wide variety of information relating to the vehicle systems, such as engine RPM, emissions control, coolant temperature, vehicle speed, timing advance, throttle position, and oxygen sensing, and many other types of data. Telematics devices installed within vehicles may be configured to access the vehicle computers and sensor data, and transmit the data to a display within the vehicle, a personal computer or mobile device, or

to a centralized data processing system. Data obtained from OBD systems has been used for a variety of purposes, including maintenance, diagnosis, and analysis.

SUMMARY

[0005] The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosure. The summary is not an extensive overview of the disclosure. It is neither intended to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the description below.

[0006] Aspects of the disclosure relate to methods, computer-readable media, and apparatuses for analyzing vehicle operation data, or driving data, and calculating or adjusting a driver score based on the analyzed driving data. One or more on-board data recording systems within a vehicle, for example, a telematics device, may be configured to collect vehicle operational data and transmit the data to a vehicle operation computer system or a driving analysis server. Vehicle operational data may include various data collected by the vehicle's internal sensors, computers, and cameras, such as the vehicle's speed, rates of acceleration or braking, instances of swerving, impacts to the vehicle, data indicating driver distractions within the vehicle, and usage of seat belts, turn signals, and other vehicle controls. Based on the vehicle operational data, the driving analysis server may be configured to identify one or more potentially high-risk or unsafe driving behaviors at a first vehicle, for example, speeding, sudden acceleration or braking, swerving, lane departures, tailgating, etc. According to certain aspects, additional data may be retrieved from one or more external data sources in order to identify the potentially high-risk or unsafe driving behaviors. For example, speeding by the first vehicle may be determined using speed sensor data and location data from the first vehicle, along with speed limit data from a separate data source. As another example, occurrences of aggressive driving in bad weather may be identified using operation data from the vehicle, along with weather data from a separate weather database.

[0007] In accordance with further aspects of the present disclosure, corresponding vehicle operation data may be retrieved for one or more additional vehicles at a similar time, location, and/or circumstances to the potentially high-risk or unsafe driving behavior of the first vehicle. For example, if the potentially high-risk or unsafe driving behavior of the first vehicle is speeding along an isolated road, then corresponding speed data may be retrieved from other vehicles along the same isolated road to identify one or more other vehicles that also may have been speeding along the same road. The driving analysis server may identify and measure the number of occurrences of the same driving behavior to determine whether or not to adjust a driver score for the first vehicle based on the driving behavior. For instance, if a high percentage of other vehicles have the same driving behavior as the first vehicle, the behavior may be deemed less risky or unsafe and the driver score associated with the first vehicle might not be lowered, whereas if a low percentage of other vehicles have the same driving behavior, the behavior may be deemed more risky or unsafe and the driver score associated with the first vehicle may be lowered.

[0008] Other features and advantages of the disclosure will be apparent from the additional description provided herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A more complete understanding of the present invention and the advantages thereof may be acquired by referring to the following description in consideration of the accompanying drawings, in which like reference numbers indicate like features, and wherein:

[0010] FIG. 1 illustrates a network environment and computing systems that may be used to implement aspects of the disclosure.

[0011] FIG. 2 is a diagram of a driving analysis system, according to one or more aspects of the disclosure.

[0012] FIG. 3 is a flow diagram illustrating an example method of adjusting a driver score based on the driving behavior of a first vehicle and one or more other vehicles, according to one or more aspects of the disclosure.

DETAILED DESCRIPTION

[0013] In the following description of the various embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration, various embodiments of the disclosure that may be practiced. It is to be understood that other embodiments may be utilized.

[0014] As will be appreciated by one of skill in the art upon reading the following disclosure, various aspects described herein may be embodied as a method, a computer system, or a computer program product. Accordingly, those aspects may take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment combining software and hardware aspects. Furthermore, such aspects may take the form of a computer program product stored by one or more computer-readable storage media having computer-readable program code, or instructions, embodied in or on the storage media. Any suitable computer readable storage media may be utilized, including hard disks, CD-ROMs, optical storage devices, magnetic storage devices, and/or any combination thereof. In addition, various signals representing data or events as described herein may be transferred between a source and a destination in the form of electromagnetic waves traveling through signal-conducting media such as metal wires, optical fibers, and/or wireless transmission media (e.g., air and/or space).

[0015] FIG. 1 illustrates a block diagram of a computing device (or system) **101** in communication system **100** that may be used according to one or more illustrative embodiments of the disclosure. The device **101** may have a processor **103** for controlling overall operation of the device **101** and its associated components, including RAM **105**, ROM **107**, input/output module **109**, and memory **115**. The computing device **101**, along with one or more additional devices (e.g., terminals **141**, **151**) may correspond to any of multiple systems or devices, such as a driving analysis server or system, configured as described herein for receiving and analyzing vehicle driving data and calculating driver scores based on the driving analysis.

[0016] Input/Output (I/O) **109** may include a microphone, keypad, touch screen, and/or stylus through which a user of the computing device **101** may provide input, and may also include one or more of a speaker for providing audio output and a video display device for providing textual, audiovisual and/or graphical output. Software may be stored within memory **115** and/or storage to provide instructions to processor **103** for enabling device **101** to perform various functions. For example, memory **115** may store software used by the device **101**, such as an operating system **117**, application programs **119**, and an associated internal database **121**. Processor **103** and its associated components may allow the driving analysis system **101** to execute a series of computer-readable instructions to receive a driving data from a first vehicle, retrieve additional driving data for other vehicles corresponding to first vehicle driving data, and perform a driving data analysis for the first vehicle.

[0017] The driving analysis system **101** may operate in a networked environment **100** supporting connections to one or more remote computers, such as terminals **141** and **151**. The terminals **141** and **151** may be personal computers, servers (e.g., web servers, database servers), or mobile communication devices (e.g., vehicle telematics devices, on-board vehicle computers, mobile phones, portable computing devices, and the like), and may include some or all of the elements described above with respect to the driving analysis system **101**. The network connections depicted in FIG. 1 include a local area network (LAN) **125** and a wide area network (WAN) **129**, and a wireless telecommunications network **133**, but may also include other networks. When used in a LAN networking environment, the driving analysis system **101** may be connected to the LAN **125**

through a network interface or adapter **123**. When used in a WAN networking environment, the system **101** may include a modem **127** or other means for establishing communications over the WAN **129**, such as network **131** (e.g., the Internet). When used in a wireless telecommunications network **133**, the system **101** may include one or more transceivers, digital signal processors, and additional circuitry and software for communicating with wireless computing devices **141** (e.g., mobile phones, vehicle telematics devices) via one or more network devices **135** (e.g., base transceiver stations) in the wireless network **133**.

[0018] It will be appreciated that the network connections shown are illustrative and other means of establishing a communications link between the computers may be used. The existence of any of various network protocols such as TCP/IP, Ethernet, FTP, HTTP and the like, and of various wireless communication technologies such as GSM, CDMA, WiFi, and WiMAX, is presumed, and the various computing devices and driving analysis system components described herein may be configured to communicate using any of these network protocols or technologies.

[0019] Additionally, one or more application programs **119** used by the driving analysis server/system **101** may include computer executable instructions (e.g., driving analysis programs and driver score algorithms) for receiving vehicle driving data, retrieving additional driving data for other vehicles, analyzing and comparing the driving data with respect to specific driving behaviors, performing a driving data analysis or driver score computation for one or more vehicles or drivers, and performing other related functions as described herein.

[0020] As used herein, a driver score (or driving score) may refer to a measurement of driving abilities, safe driving habits, and other driver information. A driver score may be a rating generated by an insurance company, financial institution, or other organization, based on the driver's age, vision, medical history, driving record, and/or other account data relating to the driver. For example, an insurance company server **101** may periodically calculate driver scores for one or more of the insurance company's customers, and may use the driver scores to perform insurance analyses and determinations (e.g., determine coverage, calculate premiums and deductibles, award safe driver discounts, etc.). As discussed below, the driver score may be increased or decreased based on the real-time data collected by on-board data recording systems (e.g., vehicle sensors, cameras, and telematics devices), and other systems for measuring driving performance. For example, if a driver consistently drives within posted speed limits, wears a seatbelt, and keeps the vehicle in good repair, the driver score may be increased. Alternatively, if a driver regularly speeds, drives aggressively, and does not properly maintain the vehicle, the driver score may be lowered. It should be understood that a driver score, as used herein, may be associated with an individual, group of individuals, or a vehicle. For instance, a family, group of friends or co-workers, or other group that shares a vehicle, may have a single driver score that is shared by the group. Additionally, a vehicle may have an associated driver score that is based on one or more primary drivers of the vehicle and can be affected by the driving behavior of any of the vehicle's drivers. In other examples, a vehicle may be configured to identify different drivers, and each driver of the vehicle may have a separate driver score.

[0021] FIG. 2 is a diagram of an illustrative driving analysis system **200**. Each component shown in FIG. 2 may be implemented in hardware, software, or a combination of the two. Additionally, each component of the driving analysis system **200** may include a computing device (or system) having some or all of the structural components described above for computing device **101**.

[0022] The driving analysis system **200** shown in FIG. 2 includes a vehicle **210**, such as an automobile, motorcycle, or other vehicle for which a driving analysis may be performed and for which a driver score may be calculated. The vehicle **210** may include one or more on-board data recording systems, for example, on-board diagnostic (ODB) systems, telematics devices **216**, and/or vehicle computer systems, which may include or may be configured to communicate with vehicle sensors **212**, proximity sensors and cameras **214**, and other on-board data detection devices.

[0023] Vehicle operation sensors **212** refer to a set of sensors and data detection devices capable of

detecting and recording various conditions at the vehicle and operational parameters of the vehicle. For example, sensors **212** may detect and store data corresponding to the vehicle's speed, distances driven, rates of acceleration or braking, and specific instances of sudden acceleration, braking, and swerving. Sensors **212** also may detect and store data received from the vehicle's **210** internal systems, such as impact to the body of the vehicle, air bag deployment, headlights usage, brake light operation, door opening and closing, door locking and unlocking, cruise control usage, hazard lights usage, windshield wiper usage, horn usage, turn signal usage, seat belt usage, phone and radio usage within the vehicle, maintenance performed on the vehicle, and other data collected by the vehicle's computer systems.

[0024] Additional sensors **212** may detect and store the external driving conditions, for example, external temperature, rain, snow, light levels, and sun position for driver visibility. Sensors **212** also may detect and store data relating to moving violations and the observance of traffic signals and signs by the vehicle **210**. Additional sensors **212** may detect and store data relating to the maintenance of the vehicle **210**, such as the engine status, oil level, engine coolant temperature, odometer reading, the level of fuel in the fuel tank, engine revolutions per minute (RPMs), and/or tire pressure.

[0025] The vehicle **210** also may include one or more cameras and proximity sensors **214** capable of recording additional conditions inside or outside of the vehicle **210**. Internal cameras **214** may detect conditions such as the number of the passengers in the vehicle **210**, and potential sources of driver distraction within the vehicle (e.g., pets, phone usage, unsecured objects in the vehicle). External cameras and proximity sensors **214** may detect other nearby vehicles, traffic levels, road conditions, traffic obstructions, animals, cyclists, pedestrians, and other conditions that may factor into a driving analysis.

[0026] The operational sensors **212** and the cameras and proximity sensors **214** may store data within the vehicle **210**, and/or may transmit the data to one or more external computer systems (e.g., a vehicle operation computer system **225** and/or a driving analysis server **220**). As shown in FIG. 2, the operation sensors **212**, and the cameras and proximity sensors **214**, may be configured to transmit data to a vehicle operation computer system **225** via a telematics device **216**. In other examples, one or more of the operation sensors **212** and/or the cameras and proximity sensors **214** may be configured to transmit data directly without using a telematics device **216**. For example, telematics device **216** may be configured to receive and transmit data from operational sensors **212**, while one or more cameras and proximity sensors **214** may be configured to directly transmit data to a vehicle operation computer system **225** or a driving analysis server **220** without using the telematics device **216**. Thus, telematics device **216** may be optional in certain embodiments where one or more sensors or cameras **212** and **214** within the vehicle **210** may be configured to independently capture, store, and transmit vehicle operation and driving data.

[0027] Telematics device **216** may be a computing device containing many or all of the hardware/software components as the computing device **101** depicted in FIG. 1. As discussed above, the telematics device **216** may receive vehicle operation and driving data from vehicle sensors **212**, and proximity sensors and cameras **214**, and may transmit the data to one or more external computer systems (e.g., a vehicle operation computer system **225** and/or a driving analysis server **220**) over a wireless transmission network. Telematics device **216** also may be configured to detect or determine additional types of data relating to real-time driving and the condition of the vehicle **210**. In certain embodiments, the telematics device **216** may contain or may be integral with one or more of the vehicle sensors **212** and proximity sensors and cameras **214** discussed above, and/or with one or more additional sensors discussed below.

[0028] Additionally, the telematics device **216** may be configured to collect data regarding the number of passengers and the types of passengers (e.g. adults, children, teenagers, pets, etc.) in the vehicle **210**. The telematics device **216** also may be configured to collect data a driver's movements or the condition of a driver. For example, the telematics device **216** may include or communicate

with sensors that monitor a driver's movements, such as the driver's eye position and/or head position, etc. Additionally, the telematics device **216** may collect data regarding the physical or mental state of the driver, such as fatigue or intoxication. The condition of the driver may be determined through the movements of the driver or through sensors, for example, sensors that detect the content of alcohol in the air or blood alcohol content of the driver, such as a breathalyzer. [0029] The telematics device **216** also may collect information regarding the driver's route choice, whether the driver follows a given route, and to classify the type of trip (e.g. commute, errand, new route, etc.). In certain embodiments, the telematics device **216** may be configured to communicate with the sensors and/or cameras **212** and **214** to determine when and how often the vehicle **210** stays in a single lane or strays into other lanes. To determine the vehicle's route, lane position, and other data, the telematics device **216** may include or may receive data from a mobile telephone, a Global Positioning System (GPS), locational sensors positioned inside a vehicle, or locational sensors or devices remote from the vehicle **210**.

[0030] The telematics device **216** also may store the type of the vehicle **210**, for example, the make, model, trim (or sub-model), year, and/or engine specifications. The vehicle type may be programmed into the telematics device **216** by a user or customer, determined by accessing a remote computer system, such as an insurance company or financial institution server, or may be determined from the vehicle itself (e.g., by accessing the vehicle's **210** computer systems).

[0031] Vehicle operation computer system **225** may be a computing device separate from the vehicle **210**, containing some or all of the hardware/software components as the computing device **101** depicted in FIG. 1. The vehicle operation computer system **225** may be configured to receive and store the vehicle operation data discussed above from vehicle **210**, and similar vehicle operation data from one or more other vehicles **210a-n**. In the example shown in FIG. 2, the vehicle operation computer system **225** includes a vehicle operation database **227** that may be configured to store the vehicle operation data collected from the vehicle sensors **212**, proximity sensors and cameras **214**, and telematics devices **216** of a plurality of vehicles. The vehicle operation database **227** may store operational sensor data, proximity sensor data, camera data (e.g., image, audio, and/or video), location data and/or time data for multiple vehicles **210**.

[0032] Data stored in the vehicle operation database **227** may be organized in any of several different manners. For example, a table in the vehicle operation database **227** may contain all of the vehicle operation data for a specific vehicle **210**, similar to a vehicle event log. Other tables in the vehicle operation database **227** may store certain types of data for multiple vehicles. For instance, tables may store specific driving behaviors (e.g., driving speed, acceleration and braking rates, swerving, tailgating, use of turn signals or other vehicle controls, etc.) for multiples vehicles **210** at specific locations, such as specific neighborhoods, roads, or intersections. Vehicle operation data may also be organized by time, so that the driving behaviors of multiples vehicles **210** may be stored or grouped by time (e.g., morning, afternoon, late night, rush hour, weekends, etc.) as well as location.

[0033] The system **200** also may include a driving analysis server **220**, containing some or all of the hardware/software components as the computing device **101** depicted in FIG. 1. The driving analysis server **220** may include hardware, software, and network components to receive vehicle operation data from the vehicle operation computer system **225** and/or directly from a plurality of vehicles **210**. The driving analysis server **220** and the vehicle operation computer system **225** may be implemented as a single server/system, or may be separate servers/systems. In some examples, the driving analysis server **220** may be a central server configured to receive vehicle operation data from a plurality of remotely located vehicle operation computer systems **225**.

[0034] As shown in FIG. 2, driving analysis server **220** may include a driving analysis module **221** and a driver score calculation module **222**. Modules **221** and **222** may be implemented in hardware and/or software configured to perform a set of specific functions within the driving analysis server **220**. For example, the driving analysis module **221** and the driver score calculation module **222**

may include one or more driving analysis/driver score calculation algorithms, which may be executed by one or more software applications running on generic or specialized hardware within the driving analysis server **220**. The driving analysis module **221** may use the vehicle operation data received from the vehicle operation computer system **225** and/or other systems to perform driving analyses for specific vehicles **210**. The driver score calculation module **222** may use the results of the driving analysis performed by module **221** to calculate or adjust a driver score for a driver of a vehicle **210** based on specific driving behaviors. Further descriptions and examples of the algorithms, functions, and analyses that may be executed by the driving analysis module **221** and the driver score calculation module **222** are described below in reference to FIG. 3.

[0035] To perform driving analyses and driver score calculations, the driving analysis server **220** may initiate communication with and/or retrieve data from one or more vehicles **210**, vehicle operation computer systems **225**, and additional computer systems **231-233** storing data that may be relevant to the driving analyses and driver score calculations. For example, one or more traffic data storage systems **231**, such as traffic databases, may store data corresponding to the amount of traffic and certain traffic characteristics (e.g., amount of traffic, average driving speed, traffic speed distribution, and numbers and types of accidents, etc.) at various specific locations and times. One or more weather data storage systems **232**, such as weather databases, may store weather data (e.g., rain, snow, sleet, hail, temperature, wind, road conditions, visibility, etc.) at different locations and different times. One or more additional driving databases/systems **233** may store additional driving data from one or more different data sources or providers which may be relevant to the driving analyses and/or driver score calculations performed by the driving analysis server **220**. Additional driving databases/systems **233** may store data regarding events such as road hazards and traffic accidents, downed trees, power outages, road construction zones, school zones, and natural disasters that may affect the driving analyses and/or driver score calculations performed by the driving analysis server **220**. As discussed below in reference to FIG. 3, the driving analysis server **220** may retrieve and use data from databases/systems **231-233** to analyze and evaluate the driving behaviors of specific vehicles **210**.

[0036] FIG. 3 is a flow diagram illustrating an example method of performing a driving analysis based on vehicle driving data. This example method may be performed by one or more computing devices (e.g. driving analysis server **220**, vehicle operation computer system **225**, and vehicle telematics device **216**) in the driving analysis system **200**.

[0037] The steps in the example method of FIG. 3 describe performing an analysis to determine whether or not to adjust a driver score in response to a potentially high-risk or unsafe driving behavior (e.g., speeding, sudden braking, swerving, tailgating, moving violations, etc.) based on similar driving behavior data from other vehicles. For instance, speeding may be considered a potentially high-risk or unsafe driving behavior, and drivers that speed frequently and excessively may receive lower driver scores than drivers that do not speed. However, in a particular flow of traffic, if most or all of the cars are speeding, then it might be safer for a driver to drive within the flow of traffic than to drive at the speed limit. Similarly, an occurrence of sudden swerving or braking by a vehicle may indicate a high-risk or unsafe driving behavior by a driver not paying attention to the road. However, if many cars on the same road at or near the same time also brake or swerve suddenly, then the swerving or braking by these cars may indicate an unsafe road condition or obstruction (e.g., an icy road, fallen tree, disabled vehicle, etc.), rather than high-risk or unsafe driving by the drivers whose vehicles were braking or swerving.

[0038] In step **301**, a driving analysis server **220** may receive vehicle operation data (or driving data) for a first vehicle **210**. As described above in reference to FIG. 2, the driving analysis server **220** may receive vehicle operation data from one or more vehicle operation computer systems **225** and/or directly from telematics devices **216** or other systems on vehicles **210**. The first vehicle driving data may correspond to sensor data collected by sensors **212**, proximity or image data collected by sensors and cameras **214**, and/or additional data collected by a telematics device **216**

or other systems within a vehicle **210**. In addition to the vehicle operation data, the driving analysis server **220** may receive location and time information corresponding to the vehicle operation data in step **301**. Vehicle location data and time data may be received from the same sources as other vehicle operation data, or may be collected by different data sources or generated by the driving analysis server **220**. For example, the driving analysis server **220** may receive vehicle operation data from a vehicle operation system **225**, and then may initiate communication with the vehicle's telematics device **216**, GPS servers, time servers, or other systems to determine the location and time that correspond to the received vehicle operation data.

[0039] In certain embodiments, telematics devices **216**, vehicle operation systems **225**, and other data sources may transmit vehicle operation data for a first vehicle **210** to the driving analysis server **220** in real-time (or near real-time). The driving analysis server **220** may be configured to receive the vehicle operation data, and then perform real-time (or near real-time) driving analyses and driver score calculations for the first vehicle **210**. In other embodiments, vehicle operation data might not be transmitted in real-time but may be sent periodically (e.g., hourly, daily, weekly, etc.) by telematics devices **216** or vehicle operation systems **225**. Periodic transmissions of vehicle operation data may include data for a single vehicle or single driver, or for multiple vehicles or drivers. The driving analysis server **220** may be configured to receive the periodic transmissions, and then to perform periodic driving analyses and driver score calculations for one or more vehicles and drivers.

[0040] In step **302**, the driving analysis server **220** may identify one or more potentially high-risk or unsafe driving behaviors within the operation data of the vehicle **210**. The driving behaviors identified in step **302** may correspond to specific occurrences or patterns of high-risk, unsafe, or illegal driving activities that have the potential to affect the driver score of the vehicle **210** or a driver of the vehicle **210**.

[0041] For certain such driving behaviors, the driving analysis server **220** may identify the driving behavior by analyzing only the vehicle operation data, for example, occurrences of sudden braking, accelerating, swerving, and tailgating. Other driving behaviors that may be identified based only on the vehicle operation data include failure to use seatbelts, phone usage while driving, loud noise levels inside the vehicle while driving (e.g., high stereo volume or passenger noises), or other distractions in the vehicle (e.g., animated passengers or pets in the vehicle, eating while driving, texting while driving, etc.). Additionally, impacts to the body of the vehicle **210** (e.g., minor accidents, driving fast over speed bumps or dips, etc.) may be identified based only on the operation data received for the vehicle **210**.

[0042] To identify other types of driving behaviors, the driving analysis server **220** may analyze the vehicle operation data, as well as additional data retrieved from one or more external data sources. For example, to identify an occurrence of speeding by the vehicle **210**, the driving analysis server **220** may receive vehicle speed data from a telematics device **216** or vehicle operation system **225**, and may receive the vehicle's location from the telematics device **216**, vehicle operation system **225**, or a separate GPS system associated with the vehicle **210**. Then, the driving analysis server **220** may access a speed limit database to determine the legal speed limit at the location of the vehicle **210**, and may compare the speed limit to the detected speed of the vehicle. The driving analysis server **220** may identify other moving violations using similar techniques. For example, the driving analysis server **220** may identify a failure to use proper turn signals by analyzing the turn signal usage of the vehicle **210**, as compared to the location/driving route of the vehicle. Stop sign violations, illegal turns, and U-turn violations may be identified by comparing the driving route of the vehicle **210** to a database of the traffic regulations and posted traffic signs at different streets and intersections along the driving route.

[0043] Additional driving behaviors that may be identified in step **302** include occurrences of risky or aggressive driving under adverse driving conditions or in safe driving areas. For example, it may be deemed high-risk or unsafe to drive a vehicle **210** at the maximum speed limit during a

rainstorm, snowstorm, or on slick or icy road conditions. To detect an occurrence of this type of driving behavior, the driving analysis server **220** may analyze the speed of the vehicle **210**, the speed limit at the vehicle's location, and the weather or road conditions at the time the vehicle was being driven at that location. Additionally, it may be deemed high-risk or unsafe to drive aggressively in safe driving areas, such as construction zones and school zones. To identify an occurrence of aggressive driving in a safe driving area, the driving analysis server **220** may analyze certain vehicle operation data (e.g., sudden acceleration or braking, phone usage while, other driver distractions, etc.), and compare the vehicle's location to a database of safe driving areas.

[0044] In step **303**, the driving analysis server **220** may retrieve driving data for one or more additional vehicles (e.g., vehicles **210a-n**) corresponding to the driving behavior(s) identified for the first vehicle **210** in step **302**. As discussed below, the data retrieved in step **303** may allow the driving analysis server **220** to identify occurrences of the same potentially high-risk or unsafe driving behavior in other vehicles at similar times, locations, and/or circumstances. For example, if the driving behavior identified in step **302** is an occurrence of speeding by the first vehicle **210** on Route 1 at 2:00 pm on Saturday, then in step **303** the driving analysis server **220** may retrieve vehicle speed data for a plurality of other vehicles **210a-n** that were driven on Route 1 at 2:00 pm on Saturday. Thus, the driving behavior identified for the first vehicle (e.g., speeding) can be compared to the corresponding driving behaviors (e.g., speeding or not speeding) for other vehicles being driven at the same time and location, and under the same circumstances.

[0045] In certain embodiments, the driving analysis server **220** may identify a potentially high-risk or unsafe driving behavior of the first vehicle **210** in real-time or near real-time (in step **302**), and may retrieve corresponding driving data for other vehicles near the first vehicle at or near the same time (in step **303**). After identifying a potentially high-risk or unsafe driving behavior in the first vehicle, one or more of the telematics devices **216** of the first vehicle, the vehicle operation system **225**, or the driving analysis server **220** may be used to identify other vehicles **210a-n** in the immediate vicinity of the first vehicle **210** at the time of the identified driving behavior. GPS systems or proximity sensors within the vehicle **210** and the vehicles **210a-n**, and/or vehicle location data from other sources, also may be used to identify the set of vehicles in the immediate vicinity of the first vehicle **210** at the time of the driving behavior.

[0046] In some instances, the driving analysis server **220** may already have, or may already be configured to receive, the corresponding data from the additional nearby vehicles **210a-n**. For example, the driving analysis server **220** may receive a stream or batch of vehicle operation data from a vehicle operation system **225** corresponding to all of the vehicles in the same geographic area at the same time. Thus, the driving analysis server **220** may receive the corresponding vehicle operation data for any nearby vehicles **210a-n** at the time same that it receives the data for the first vehicle **210** in step **301**. In other instances, the driving analysis server **220** may specifically request the information from the vehicle operation system **225** or from vehicle telematics devices **216**, for any vehicles determined to be nearby the first vehicle **210** at the time of the identified driving behavior. Thus, the driving analysis server **220** may be configured to locate, initiate communication with, and request specific sensor/camera data from a vehicle operation system **225** or from one or more vehicle telematics devices **216** corresponding to the nearby vehicles **210a-n**. For example, if the potentially high-risk or unsafe driving behavior is tailgating by the first vehicle **210**, then the driving analysis server **220** may identify other nearby vehicles **210a-n** at the time of the identified tailgating, and may specifically request data from the proximity sensors and external cameras **214** of the other vehicles **210a-n** that may be used to identify similar occurrences of tailgating by the other vehicles **210a-n**.

[0047] Depending on the time and location of the potentially high-risk or unsafe driving behavior by the first vehicle **210**, the driving analysis server **220** may be unable to locate a significant number of additional nearby vehicles, or may be unable retrieve corresponding driving data from the nearby vehicles. For example, driving behaviors identified in step **302** may take place on

remote and isolated roads, or nearby vehicles might not have the necessary sensors **212-214** and telematics devices **216** to detect and transmit the corresponding driving data to the driving analysis server **220**. In such cases, the driving analysis server **220** may retrieve corresponding driving data from one or more other data sources. For example, the driving analysis server **220** may request corresponding traffic data from traffic databases **231**, including the amount of traffic, average driving speed, lane changes, and instances of accidents at times and locations corresponding to the driving behavior of the first vehicle **210**.

[0048] When there is insufficient driving data available for other vehicles **210a-n** at the time and location corresponding to the driving behavior of the first vehicle **210**, the driving analysis server **220** may retrieve similar data for different times and/or different locations that are similar to the time and location of the first vehicle's driving behavior. The specific type of data retrieved by the driving analysis server **220** (e.g., the other vehicles **210a-n** identified, the specific sensor data used and other vehicle operation data retrieved, and the times and locations corresponding to the vehicle operation data) may depend on the type of driving behavior identified for the first vehicle **210** in step **302**, and other specific circumstances relating to the driving behavior.

[0049] For example, if the driving behavior identified for the first vehicle **210** in step **302** is speeding along a particular section of Route 1 at 2:00 am on Monday, then in step **303**, the driving analysis server **220** may retrieve corresponding vehicle speed data for any vehicles on the same section of Route 1 between 12:00 am and 4:00 am, Monday through Friday. Depending on the amount of data available, speed data during this time window may be collected over a number of previous weeks or months to provide an adequate sample size. As another example, if the driving behavior identified for the first vehicle **210** in step **302** is failure to stop completely at a stop sign at 2:00 pm on Tuesday afternoon, then in step **303**, the driving analysis server **220** may retrieve corresponding vehicle data (e.g., driving speed and braking data) for any vehicles at the same intersection at any time in the previous several months. As another example, if the driving behavior identified for the first vehicle **210** in step **302** is a sudden swerve at a certain point on Main Street at 4:30 pm on Wednesday, then in step **303**, the driving analysis server **220** may retrieve corresponding vehicle data (e.g., steering/swerving data) only for vehicles on the same section of Main Street, and only between 3:30 pm and 5:30 pm on the same Wednesday afternoon. The sudden swerve in this example may be due to an accident, disabled vehicle, or temporary road obstruction, and therefore data might not be collected for previous weeks and months because such data might not accurately reflect the similar circumstances to the driving behavior identified for the first vehicle **210**. As yet another example, if the driving behavior identified for the first vehicle **210** in step **302** is driving aggressively during a rain or snow storm on Thursday morning, then in step **303**, the driving analysis server **220** may retrieve corresponding vehicle data (e.g., speed, acceleration and braking rates, steering/swerving data) for vehicles within the same general area as the rain or storm at any time during the storm.

[0050] In step **304**, the driving analysis server **220** may use the additional vehicle driving data retrieved in step **303** to identify occurrences of the same or similar driving behaviors as the first vehicle driving behavior that was identified in step **302**. The driving analysis server **220** may perform the analysis in step **304** separately for each other vehicle and for each driving behavior. That is, for each additional vehicle that vehicle driving data was retrieved in step **303**, the driving analysis server **220** may analyze the driving data for that vehicle to determine if one or more instances occur for the same (or a similar) driving behavior. For example, if the driving behavior identified for the first vehicle **210** in step **302** is speeding by greater than 20 miles-per-hour (MPH) along a particular section of Main Street, then in step **304**, the driving analysis server **220** may analyze each trip by each other vehicle down the same section of Main Street to identify other occurrences of speeding greater than 20 MPH. If the data retrieved in step **303** includes vehicle driving data for 500 vehicle trips down the same section of Main Street, then the driving analysis server **220** may analyze the vehicle's speed data on each trip and classify each of the 500 trips as

either an occurrence of speeding greater than 20 MPH or an occurrence of not speeding greater than 20 MPH.

[0051] When identifying occurrences of the same or similar driving behaviors, between the first vehicle **210** and the additional vehicles **210a-210n**, the driving analysis server **220** may compare the specific vehicle operational data (e.g., vehicle speeds, rates of accelerating and braking, severity of swerving, tailgating distances, number and severity of lane departures, etc.) between the vehicles. In certain instances, thresholds may be defined for each driving behavior (e.g., speeding, sudden acceleration and braking, swerving, tailgating, other moving violations, etc.) so that occurrences and non-occurrences of the driving behavior can be identified within the data retrieved in step **303**. The thresholds may depend on the general type of driving behavior as well as specific situational factors. For example, an occurrence of speeding for one analysis in step **304** may be defined 10 miles-per-hour (MPH) over the posted speed limit, whereas in a different analysis in step **304** (e.g., on a different road, with different road conditions, and at a different time of day) may define speed as 5 MPH over the posted limit. The thresholds for occurrences of sudden accelerating, braking, and swerving, and failing to use turn signals (or other vehicle controls), may be based on the readings of the vehicle's operational sensors **212**. The thresholds for occurrences of tailgating may be based on the vehicle's proximity sensors and cameras **214**. For other potentially high-risk or unsafe driving behaviors, the appropriate sensor data or other data sources may be used to set and use thresholds for measuring occurrences and non-occurrences of the behavior by other vehicles **210a-n**.

[0052] The data retrieved in step **303** and analyzed in step **304** may correspond to the same location and time as the identified driving behavior of the first vehicle **210** (e.g., Main Street at 1st Avenue, 1:00 pm on Monday), or may correspond to different locations (e.g., Broad Street at 1st Avenue, 1:00 pm on Monday) and/or different times (Main Street at 1st Avenue, 1:00 pm on Tuesday). For each of these examples, the driving analysis server **220** may identify occurrence or a non-occurrence of the same driving behavior identified in step **302**. However, the thresholds for identifying an occurrence or a non-occurrence may be adjusted when the analyzed data corresponds to different times and/or locations. For example, if the first vehicle **210** was speeding down a quiet isolated road at night, and the data analyzed in step **304** includes vehicle speed data for the same road during both daytime at nighttime, then the driving analysis server **220** may set a higher MPH threshold for identifying an occurrence of speeding during the daytime because of the increased daytime visibility.

[0053] In step **305**, the driving analysis server **220** determines if the number (or percentage) of occurrences of the driving behavior identified in step **304** exceeds a threshold number (or threshold percentage) of occurrences. If the number/percentage does not exceed the threshold number/percentage (**305**: No), this indicates a relatively small number of occurrences of the potentially high-risk or unsafe driving behavior among the data retrieved in step **303**. In this case, the potentially high-risk or unsafe driving behavior identified for the first vehicle **210** in step **302** is a relatively uncommon driving behavior. Accordingly, in step **306**, the driving analysis server **220** may adjust (e.g., lower) a driver score associated with the first vehicle or a driver of the first vehicle, based on the potentially high-risk or unsafe driving behavior identified in step **302**. On the other hand, if the number/percentage exceeds the threshold number/percentage (**305**: Yes), this indicates a relatively large number of occurrences of the potentially high-risk or unsafe driving behavior among the data retrieved in step **303**. In this case, the driving behavior identified for the first vehicle **210** in step **302** is a relatively common driving behavior, and may be less likely to be a high-risk or unsafe driving behavior. Accordingly, in step **307**, the driving analysis server **220** might not adjust the driver score associated with the first vehicle or driver of the first vehicle, based on the driving behavior identified in step **302**.

[0054] As an example, if the potentially high-risk or unsafe driving behavior identified for the first vehicle **210** in step **302** is speeding by greater than 10 MPH, then in step **305**, the driving analysis

server **220** may compare the percentage of other vehicles that were also speeding by greater than 10 MPH against a percentage threshold (e.g., 50%). In this example, if 90% of the other vehicles (for which data was retrieved in step **303**) were also speeding by greater than 10 MPH, then the driver score associated with the first vehicle **210** will not be adjusted (after comparing the 90% occurrence rate to the 50% threshold) because the driving analysis server **220** may conclude that the same driving behavior is common among other vehicles. On the other hand, if only 5% of the other vehicles were speeding by greater than 10 MPH, then the driver score will be adjusted (after comparing the 5% occurrence rate to the 50% threshold), because the driving analysis server **220** may conclude that the first vehicle **210** was speeding excessively in comparison with other vehicles. Similar analysis and threshold comparisons may be performed for other types of potentially high-risk or unsafe driving behavior, such as sudden acceleration or braking, swerving, tailgating, lane departures, failure to use turn signals or other vehicle controls, or other moving violations. In these examples, if a relatively high percentage of other vehicles have the same driving behavior, the behavior may be deemed less risky or unsafe and the driver score associated with the first vehicle **210** might not be lowered, or it may be lowered by less than it otherwise would be. If a relatively low percentage of other vehicles have the same driving behavior, the behavior may be deemed more risky or unsafe and the driver score associated with the first vehicle **210** may be lowered, or it may be lowered by more than it otherwise would be.

[0055] In certain examples, the driving analysis server **220**, vehicle operation computer system **225**, and/or the telematics device **216** may be configured to send data to one or more output devices that may be visible to the driver or other users. For example, the telematics device **216** or vehicle operation computer system **225** may send vehicle operation data and any identified occurrences of potentially high-risk or unsafe driving behavior to a separate computing device (e.g., personal computer, e-mail account, mobile device) associated with the drivers or the owners of the vehicle **210**. Additionally, the driving analysis server **220** may be configured to notify vehicle drivers or owners of any adjustments made to the driver's/vehicle's driver score in step **306**. Any occurrences of identified driving behaviors that may affect the driver score and/or any adjustments to the driver score also may be displayed within the vehicle **210**. For example, driving analysis server **220**, vehicle operation computer system **225**, and/or the telematics device **216** may be configured to project a warning of a potentially high-risk or unsafe driving behavior, or an updated driver score, on the display console, windshield, or rear view mirror of the vehicle **210**.

[0056] As discussed above in step **304**, the driving analysis server **220** may analyze the corresponding vehicle driving data in step **304** and may classify each vehicle and driving behavior as either an occurrence or a non-occurrence of the first vehicle driving behavior identified in step **302**. However, in other examples, the driving analysis server **220** need not classify each vehicle/driving behavior as an occurrence or a non-occurrence, but instead may perform a statistical analysis on the complete set of corresponding vehicle driving data. In such examples, driving analysis server **220** may determine a statistical distribution (or statistical function) representing the driving behavior, and then may place the first vehicle driving behavior within the statistical distribution or function. For instance, after determining a statistical distribution for the driving behavior, the driving analysis server **220** may determine that the identified behavior of the first vehicle **210** corresponds to the Nth percentile among the corresponding vehicle driving data retrieved in step **303**. In these examples, the threshold in step **305** may correspond to a percentile threshold (e.g., 30th percentile 50th percentile, 75th percentile, 98th percentile, etc.) for the driving behavior of the first vehicle **210**. If the magnitude or severity of the potentially high-risk or unsafe driving behavior is greater than the percentile threshold, then the driver score for the vehicle/driver may be adjusted (step **306**), and if the magnitude or severity of the driving behavior is not greater than the percentile threshold, then the driver score for the vehicle/driver may not be adjusted (step **307**).

[0057] While the aspects described herein have been discussed with respect to specific examples

including various modes of carrying out aspects of the disclosure, those skilled in the art will appreciate that there are numerous variations and permutations of the above described systems and techniques that fall within the spirit and scope of the invention.

Claims

1. (canceled)
2. A driving analysis system comprising: at least one processor; and a memory storing computer-executable instructions, which when executed by the at least one processor, cause the driving analysis system to: generate a driving score for a vehicle; receive vehicle driving data associated with the vehicle and captured using one or more vehicle operation sensors; identify an occurrence of a driving behavior in the vehicle driving data, the driving behavior including at least one of sudden acceleration, sudden braking, speeding, swerving, an impact to the vehicle, tailgating, failing to use a vehicle control, lane departure, or a moving violation; retrieve driving data corresponding to the vehicle driving data stored in one or more data storage systems; analyze the driving data to identify an occurrence of the driving behavior in the driving data; generate an updated driver score of the vehicle by adjusting the driving score based on whether the occurrence of the driving behavior in the driving data is identified; generate an output based on whether the occurrence of the driving behavior in the driving data is identified; and transmit the output to a computing device to cause the computing device to indicate the output.
3. The driving analysis system of claim 2, wherein the driving data includes at least one of sensor data, proximity data, image data, or telematics data.
4. The driving analysis system of claim 2, wherein the computer-executable instructions, which when executed by the at least one processor, further cause the driving analysis system to: determine a time and a location associated with the vehicle driving data.
5. The driving analysis system of claim 4, wherein the driving data is retrieved from the one or more data storage systems using at least one of the time or the location.
6. The driving analysis system of claim 2, wherein the one or more vehicle operation sensors are configured to detect at least one of speed, distance driven, rate of acceleration, rate of braking, sudden acceleration, sudden braking, or swerving.
7. The driving analysis system of claim 2, wherein the vehicle driving data is received from a vehicle on-board data recording system.
8. The driving analysis system of claim 2, wherein the output includes an indication of at least one of the occurrence of the driving behavior or the updated driver score.
9. A method comprising: generating a driving score for a vehicle; receiving vehicle driving data associated with the vehicle and captured using one or more sensors; determining an occurrence of a driving behavior in the vehicle driving data, the driving behavior including at least one of sudden acceleration, sudden braking, speeding, swerving, an impact to the vehicle, tailgating, failing to use a vehicle control, lane departure, or a moving violation; retrieving driving data corresponding to the vehicle driving data from one or more data storage systems; analyzing the driving data to identify an occurrence of the driving behavior in the driving data; generating an updated driver score of the vehicle by adjusting the driving score based on whether the occurrence of the driving behavior in the driving data is identified; generating an output based on whether the occurrence of the driving behavior in the driving data is identified; and transmitting the output to a computing device to cause the computing device to indicate the output.
10. The method of claim 9, wherein the driving data includes at least one of sensor data, proximity data, image data, or telematics data.
11. The method of claim 9, further comprising: determining a time and a location associated with the vehicle driving data.
12. The method of claim 11, wherein the driving data is retrieved from the one or more data storage

systems using at least one of the time or the location.

13. The method of claim 9, wherein the one or more sensors are configured to detect at least one of speed, distance driven, rate of acceleration, rate of braking, sudden acceleration, sudden braking, or swerving.

14. The method of claim 9, wherein the vehicle driving data is received from a vehicle on-board data recording system.

15. The method of claim 9, wherein the output includes an indication of at least one of the occurrence of the driving behavior or the updated driver score.

16. One or more non-transitory computer-readable media storing computer executable instructions that, when executed by a processor, cause a computing device to: generate a driving score for a vehicle; receive vehicle driving data associated with the vehicle and captured using one or more vehicle operation sensors; identify an occurrence of a driving behavior in the vehicle driving data, the driving behavior including at least one of sudden acceleration, sudden braking, speeding, swerving, an impact to the vehicle, tailgating, failing to use a vehicle control, lane departure, or a moving violation; retrieve driving data corresponding to the vehicle driving data stored in one or more data storage systems; analyze the driving data to identify an occurrence of the driving behavior in the driving data; generate an updated driver score of the vehicle by adjusting the driving score based on whether the occurrence of the driving behavior in the driving data is identified; generate an output based on whether the occurrence of the driving behavior in the driving data is identified; and transmit the output to a computing device to cause the computing device to indicate the output.

17. The one or more non-transitory computer-readable media of claim 16, wherein the driving data includes at least one of sensor data, proximity data, image data, or telematics data.

18. The one or more non-transitory computer-readable media of claim 16, wherein the driving data is retrieved from the one or more data storage systems using at least one of a time or a location associated with the vehicle driving data.

19. The one or more non-transitory computer-readable media of claim 16, wherein the one or more vehicle operation sensors are configured to detect at least one of speed, distance driven, rate of acceleration, rate of braking, sudden acceleration, sudden braking, or swerving.

20. The one or more non-transitory computer-readable media of claim 16, wherein the vehicle driving data is received from a vehicle on-board data recording system.

21. The one or more non-transitory computer-readable media of claim 16, wherein the output includes an indication of at least one of the occurrence of the driving behavior or the updated driver score.
